GRANT NEILSON

Bioinformatician with seven years of industry experience analyzing multiple NGS data types, with a specialism in single cell RNA & whole genome sequencing. Highly proficient in R and Python, with a strong track record of developing analytical tools and packages for high-dimensional omics data. Passionate about building robust analytical workflows and contributing to open-source repositories to drive forward scientific research. An articulate and reliable individual with excellent communication skills and a strong track record of effective collaboration across multidisciplinary teams.

View this CV online

INDUSTRY EXPERIENCE

2026 | 2023

Senior Computational Biologist

CoSyne Therapeutics — London, UK

- Designed and implemented the end-to-end bioinformatic analysis for high-throughput single cell mosaic drug screening, deploying hyperscale cloud infrastructure to run alignment and quality-control pipelines for ~3 million cells. Designed and implemented scalable analysis tools (bixverse, contributor) for cell demultiplexing, sensitivity profiling of cell lines, transcriptomic stratification and multi-omic biomarker identification.
- Drove biomarker discovery in glioblastoma by conducting WGS analysis on 80 primary tumour-derived cell lines, delivering comprehensive genetic characterisation (SNVs, SVs, CNVs). Integrated genomic profiles with CRISPRi screen data using advanced statistical models to uncover novel genetic dependencies and biomarker-target pairs.
- Built a target prioritisation framework that integrated high-throughput CRISPRi phenotypic screen data with public evidence from OpenTargets, enabling systematic ranking of candidate targets to feed the discovery pipeline.
- Co-developed an agentic hypothesis-generation platform that integrates CoSyne's internal omics data with public evidence from OpenTargets, ChEMBL, DepMap and ClinicalTrials.gov. Exposed to Claude as a set of MCP tools, agents plan and execute their own multi-step queries across both, chaining rare-variant genetic evidence to CRISPR dependencies and drug tractability to propose and rank novel target hypotheses for experimental follow-up.
- Owned ingestion into the company DuckLake lakehouse, curating 70+ schema-validated tables that unify internal RNA-seq, WGS, methylation, CRISPR and drug-response data with public resources including OpenTargets, DepMap NextGen 3D screens, CCLE/PRISM and Orphanet, creating a single queryable evidence base for target identification.
- Executed single-cell CNV (sc-CNV) calling on proprietary perturb scRNA-seq data, identifying sub-clonal populations and contributing to the target identification platform.
- Internal R&D: Responsible for developing an internal container system using Docker and ECR. Author and Lead developer on several internal R and Python packages with responsibilities for maintenance. Established best practices for code development and review. Designed the templating system for all internal projects & packages using cruft.

CONTACT

grant.neilson5@gmail.com
GitHub
LinkedIn

LANGUAGES

R
Python
Bash
Nextflow
SQL

TECHNOLOGIES

GitLab/GitHub
Docker
AWS
Azure
DuckDB
MCP/LLM Agents

OPEN SOURCE & PUBLIC PROJECTS

bixverse
manifoldsR
cosyne-cnvutils
Sarek

*Made with the R package **pagedown**.*

The source code is available on GitHub.

Last updated October 2026.

2023 | 2020

Computational Biologist

Fios Genomics — Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

2020 | 2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School — Exeter, UK

- Developed novel bioinformatics pipelines for Epigenome/Genome wide association studies, mQTL analysis and quality control of Illumina EPIC array data.
- Took the lead on a project performing EWAS and GWAS analysis in ADHD patients. This involved fitting linear and logistic regression models while accounting for covariates.
- Was selected for a secondment to New York to work at the Mount Sinai medical school as a visiting bioinformatician (cancelled due to COVID-19).

EDUCATION

2019 | 2015

MSci Biology

University of Southampton — Southampton, UK

- Grade : First Class Honours
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015 | 2013

School

Merchiston Castle School — Edinburgh, UK

- A Levels: Biology, Chemistry, Maths

SELECTED PUBLICATIONS

2025

TP53 and RB1 are predictive genetic biomarkers for sensitivity to cytarabine in gliomas

BioRxiv

- L. Laaniste, J. Luginbuehl, M. Trinh, R. Watkins, G. Lueg, **G. Neilson**, et al.
- doi: https://doi.org/10.1101/2025.11.20.688938